

第三章 第二节 定积分

一、选择题

1. 答案: A

解析: 定积分 $\int_1^e t^2 \ln t dt$ 是常数, 常数的导数为 0, 即 $\frac{d}{dx} \int_1^e t^2 \ln t dt = \left(\int_1^e t^2 \ln t dt \right)' = 0$,

故选 A.

2. 答案: D

解析: 由定积分的性质可得, $\int_a^b f(x) dx = -\int_b^a f(x) dx$, 故选 D.

3. 答案: D

解析: 原式 $= \frac{1}{3} \int_3^9 f'(3x) d(3x) = \frac{1}{3} f(3x) \Big|_1^3 = \frac{1}{3} [f(9) - f(3)]$, 故选 D.

4. 答案: D

解析: 定积分的性质, “对称区间内, 偶倍奇零”

选项 D: 对称区间, 且 $x + \sin x$ 为奇函数, 所以 $\int_{-2}^2 (x + \sin x) dx = 0$, 故选 D;

选项 A: $\int_{-2}^2 x \sin x dx = 2 \int_0^2 x \sin x dx = 2 \int_0^2 x d(-\cos x) = 2x(-\cos x) \Big|_0^2 - 2 \int_0^2 1 \cdot (-\cos x) dx$
 $= -4 \cos 2 + 2 \sin x \Big|_0^2 = 2 \sin 2 - 4 \cos 2$;

选项 B: $\int_0^2 x \cos x dx = \int_0^2 x d \sin x = x \sin x \Big|_0^2 - \int_0^2 1 \cdot \sin x dx = 2 \sin 2 + \cos x \Big|_0^2 = 2 \sin 2 + \cos 2 - 1$;

选项 C: $\int_{-2}^2 (e^x + x) dx = \left(e^x + \frac{1}{2} x^2 \right) \Big|_{-2}^2 = \left(e^2 + \frac{1}{2} \times 2^2 \right) - \left(e^{-2} + \frac{1}{2} \times (-2)^2 \right) = e^2 + 2 - e^{-2} - 2 = e^2 - \frac{1}{e^2}$.

5. 答案: D

解析: 定积分的性质, “对称区间内, 偶倍奇零”. 被积函数 $\frac{x^2 \arcsin x}{x^2 + 1}$ 是奇函数, 积分区间 $[-1, 1]$ 关于

原点对称, 奇函数在对称区间定积分为 0, 故选 D.

6. 答案: A

解析: 由定积分的几何意义可得, $\int_0^{\sqrt{2}} \sqrt{2-x^2} dx = \frac{1}{4} \cdot \pi \cdot (\sqrt{2})^2 = \frac{\pi}{2}$, 选 A.

7. 答案: D

解析: 由 $\int_{-1}^1 3f(x)dx = 18$ 得 $\int_{-1}^1 f(x)dx = 6$; $\int_1^3 f(x)dx = \int_{-1}^3 f(x)dx - \int_{-1}^1 f(x)dx = 4 - 6 = -2$;
 $\int_3^{-1} g(x)dx = -\int_{-1}^3 g(x)dx = -3$; $\int_{-1}^3 \frac{1}{5}[4f(x) + 3g(x)]dx = \frac{1}{5}[4 \times 4 + 3 \times 3] = \frac{25}{5} = 5 \neq 4$, D 选项计算错误, 故选 D.

8. 答案: A

解析: 原式 $= \int_e^{+\infty} \frac{1}{(\ln x)^2} (\ln x)' dx = -\frac{1}{\ln x} \Big|_e^{+\infty} = \lim_{x \rightarrow +\infty} -\frac{1}{\ln x} + \frac{1}{\ln e} = 0 + 1 = 1$, 故选 A.

9. 答案: B

解析:

B 选项: $\int_1^{+\infty} \frac{1}{\sqrt{x^3}} dx = \int_1^{+\infty} x^{-\frac{3}{2}} dx = -2x^{-\frac{1}{2}} \Big|_1^{+\infty} = \lim_{x \rightarrow +\infty} \left(-2x^{-\frac{1}{2}} \right) + 2 = 0 + 2 = 2$, 该积分收敛, 故选 B;

A 选项: $\int_0^1 \frac{1}{\sqrt{x^3}} dx = \int_0^1 x^{-\frac{3}{2}} dx = -2x^{-\frac{1}{2}} \Big|_0^1 = 2 + \lim_{x \rightarrow 0^+} \left(-2\frac{1}{\sqrt{x}} \right) = \infty$, 故该积分发散;

C 选项: $\int_0^{+\infty} \cos x dx = \sin x \Big|_0^{+\infty} = \lim_{x \rightarrow +\infty} \sin x - \sin 0$, $\lim_{x \rightarrow +\infty} \sin x$ 极限不存在, 该反常积分发散;

D 选项: $\int_{-1}^1 \frac{1}{x^2} dx = \int_{-1}^0 x^{-2} dx + \int_0^1 x^{-2} dx$, 其中 $\int_{-1}^0 x^{-2} dx = -x^{-1} \Big|_{-1}^0 = \lim_{x \rightarrow 0^-} \left(-\frac{1}{x} \right) - 1 = +\infty$,

$\int_0^1 x^{-2} dx = -x^{-1} \Big|_0^1 = -1 - \lim_{x \rightarrow 0^+} \left(-\frac{1}{x} \right) = +\infty$, 极限不存在, 该积分发散.

10. 答案: C

解析：原式 $= \int_0^{\frac{\pi}{4}} \tan x \, d \tan x = \frac{1}{2} \tan^2 x \Big|_0^{\frac{\pi}{4}} = \frac{1}{2}$ ，故选 C.

11. 答案：D

解析：原式 $= \int_0^1 (e^x - 1)^3 d(e^x - 1) = \frac{1}{4} (e^x - 1)^4 \Big|_0^1 = \frac{1}{4} (e - 1)^4$ ，故选 D.

12. 答案：B

解析：原式 $= \left(x \int_x^{x^2} \sin t \, dt \right)' = 1 \cdot \int_x^{x^2} \sin t \, dt + x(2x \sin x^2 - 1 \cdot \sin x)$ ，故选 B.

13. 答案：A

解析：原式 $= \lim_{x \rightarrow 0} \frac{\ln(1+x)}{1} = \ln 1 = 0$ ，故选 A.

14. 答案：B

解析：令 $A = \int_0^2 f(x) \, dx$ ，则 $f(x) = x + A$ ，两边同取 $[0, 2]$ 上的积分： $\int_0^2 f(x) \, dx = \int_0^2 (x + A) \, dx$

即 $A = \left(\frac{1}{2} x^2 + Ax \right) \Big|_0^2$ ， $A = 2 + 2A$ ，解得 $A = -2$ ，所以 $f(x) = x - 2$ ，故选 B.

15. 答案：B

解析：由定积分的性质可得，“对称区间内，偶倍奇 0”.

B 选项： $\cos x$ 为偶函数，奇函数 $\times$ 偶函数 = 奇函数，故 $\int_{-1}^1 f(x) \cos x \, dx = 0$.

A 选项： $f^2(-x) = [f(-x)]^2 = (-f(x))^2 = f^2(x)$ ， $f^2(x)$ 是偶函数， $\int_{-1}^1 f^2(x) \, dx = 2 \int_0^1 f^2(x) \, dx \neq 0$.

C 选项：原式 $= \int_{-1}^1 f(x) \, dx + \int_{-1}^1 x^2 \, dx = 0 + 2 \int_0^1 x^2 \, dx = \frac{2}{3} \neq 0$.

D 选项：令 $h(x) = f(x)e^x$ ， $h(-x) = f(-x)e^{-x} = -f(x)e^{-x}$ ，既不满足奇函数也不满足偶函数，故

$\int_{-1}^1 f(x)e^x \, dx$ 未必为 0.

16. 答案: B

解析: 由定积分性质, $\int_a^b f(x)dx = \int_a^b g(x)dx$, 移项可得 $\int_a^b f(x)dx - \int_a^b g(x)dx = 0$,
 $\int_a^b [f(x) - g(x)]dx = 0$, 故选 B.

17. 答案: C

解析: 令 $x+1=t$, 则 $x=t-1$, 当 $x=-2$ 时 $t=-1$, $x=1$ 时 $t=2$,

原式 $= \int_{-1}^2 f(t)dt = \int_{-1}^0 tdt + \int_0^2 e^t dt = \frac{1}{2}t^2 \Big|_{-1}^0 + e^t \Big|_0^2 = -\frac{1}{2} + e^2 - 1 = e^2 - \frac{3}{2}$, 故选 C.

二、填空题

1. 答案: 1

解析: 原式 $= \int_0^1 (1-x)dx + \int_1^2 (x-1)dx = \left(x - \frac{1}{2}x^2\right) \Big|_0^1 + \left(\frac{1}{2}x^2 - x\right) \Big|_1^2 = \frac{1}{2} + \frac{1}{2} = 1$.

2. 答案: $x^2 + y^2 = 1$ 在 $[-1, 1]$ 上的上半圆的面积; $\frac{\pi}{2}$

解析: $x^2 + y^2 = 1$ 在 $[-1, 1]$ 上的上半圆的面积, $\int_{-1}^1 f(x)dx = \frac{1}{2}\pi \cdot 1^2 = \frac{\pi}{2}$.

3. 答案: 4

解析: 由“对称区间内, 偶倍奇 0”可得, $\int_{-1}^1 f(x)dx = 2\int_0^1 f(x)dx = 2 \times 2 = 4$.

4. 答案: $\frac{5\pi}{32}$

解析: 由华里士公式可得, $n=6$ 为偶数, 故 $\int_0^{\frac{\pi}{2}} \sin^6 x dx = \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{5\pi}{32}$.

5. 答案: $3x^2 f(x^3)dx$

解析：由变限积分函数求导可得， $\mathrm{d}\int_0^{x^3} f(t)\mathrm{d}t = \left(\int_0^{x^3} f(t)\mathrm{d}t\right)' \mathrm{d}x = f(x^3) \cdot 3x^2 \mathrm{d}x$ 。

6. 答案： $2x \cos x^2$

解析： $F'(x) = \left(\int_0^{x^2} \cos t \mathrm{d}t\right)' = \cos x^2 \cdot (x^2)' - 0 = 2x \cos x^2$ 。

7. 答案： -4

解析： $F'(x) = \left(\int_{2\sin x}^{\cos x} f(t)\mathrm{d}t\right)' = f(\cos x) \cdot (-\sin x) - f(2\sin x) \cdot 2\cos x$ ，代入 $x=0$ 得

$$F'(0) = 0 - 2 \times 1 \times f(0) = -2 \times 2 = -4.$$

8. 答案： $\frac{1}{3}$

解析：原式 $= \lim_{x \rightarrow 0} \frac{\ln(1+x^3)}{3x^2} = \lim_{x \rightarrow 0} \frac{x^3}{3x^3} = \frac{1}{3}$ 。

9. 答案： 6π

解析：由“对称区间内，偶倍奇 0”可得，原式 $= \int_{-2}^2 3\sqrt{4-x^2} \mathrm{d}x + \int_{-2}^2 x\sqrt{4-x^2} \mathrm{d}x$
 $= 6 \int_{-2}^2 \sqrt{4-x^2} \mathrm{d}x + 0 = 6 \times \frac{1}{4} \pi \times 2^2 = 6\pi$ 。

10. 答案： 2

解析：原式 $= \int_{-2}^2 \frac{1}{e^x + 1} (e^x + 1)' \mathrm{d}x = \int_{-2}^2 \frac{1}{e^x + 1} \mathrm{d}(e^x + 1) = \ln(e^x + 1) \Big|_{-2}^2 = \ln(e^2 + 1) - \ln(e^{-2} + 1)$
 $= \ln \frac{e^2 + 1}{e^{-2} + 1} = \ln \frac{e^2(1 + e^2)}{1 + e^2} = \ln e^2 = 2$ 。

三、计算题

1. (1) 答案： $\frac{3\pi}{8}$

解析：原式 $= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^4 x dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x \cos^3 x dx = 2 \int_0^{\frac{\pi}{2}} \sin^4 x dx + 0 = 2 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3\pi}{8}$.

(2) 答案：2

解析：原式 $= \int_{-1}^1 (1 + 2x\sqrt{1-x^2}) dx = \int_{-1}^1 1 dx + \int_{-1}^1 2x\sqrt{1-x^2} dx = 2 \int_0^1 1 dx + 0 = 2x \Big|_0^1 = 2$.

(3) 答案： $\frac{4}{15}$

解析：原式 $= \left(\frac{3}{5}x^5 + \frac{2}{3}x^3 - x \right) \Big|_0^1 = \frac{3}{5} + \frac{2}{3} - 1 = \frac{4}{15}$.

(4) 答案： $1 - \frac{3\pi}{4}$

解析：原式 $= \int_0^1 \left(1 - \frac{3}{1+x^2} \right) dx = (x - 3 \arctan x) \Big|_0^1 = 1 - \frac{3\pi}{4}$.

(5) 答案： $\frac{\pi^2}{8}$

解析：原式 $= \int_0^1 \arcsin x d \arcsin x = \frac{1}{2} (\arcsin x)^2 \Big|_0^1 = \frac{1}{2} \cdot \left(\frac{\pi}{2} \right)^2 = \frac{\pi^2}{8}$.

(6) 答案： $\cos \frac{2}{3}$

解析：原式 $= - \int_{\frac{\pi}{2}}^{\frac{3}{2}} \sin \frac{1}{x} \cdot \left(\frac{1}{x} \right)' dx = - \int_{\frac{\pi}{2}}^{\frac{3}{2}} \sin \frac{1}{x} d \left(\frac{1}{x} \right) = \cos \frac{1}{x} \Big|_{\frac{\pi}{2}}^{\frac{3}{2}} = \cos \frac{2}{3} - \cos \frac{\pi}{2} = \cos \frac{2}{3}$.

(7) 答案： $\frac{\pi}{2}$

解析：原式 $= \int_1^e \frac{1}{\sqrt{1-(\ln x)^2}} (\ln x)' dx = \int_1^e \frac{1}{\sqrt{1-(\ln x)^2}} d(\ln x) = \arcsin(\ln x) \Big|_1^e = \arcsin 1 - \arcsin 0$

$$= \frac{\pi}{2}.$$

(8) 答案: $\frac{1}{6}$

解析: 令 $\sqrt{5-4x}=t$, $x=\frac{5-t^2}{4}$, $dx=-\frac{t}{2}dt$, $x=-1$ 时 $t=3$, $x=1$ 时 $t=1$,

$$\text{原式} = \int_3^1 \frac{5-t^2}{t} \cdot \left(-\frac{t}{2}\right) dt = \frac{1}{8} \int_1^3 (5-t^2) dt = \frac{1}{8} \left(5t - \frac{1}{3}t^3\right) \Big|_1^3 = \frac{1}{6}.$$

(9) 答案: $\frac{4\sqrt{2}-5}{3}$

解析: 令 $\sqrt{1+x}=t$, $x=t^2-1$, $dx=2tdt$, 当 $x=0$ 时 $t=1$, 当 $x=1$ 时 $t=\sqrt{2}$,

$$\text{原式} = \int_1^{\sqrt{2}} \frac{t^2-1}{1+t} \cdot 2tdt = 2 \int_1^{\sqrt{2}} (t-1)t dt = \left(\frac{2}{3}t^3 - t^2\right) \Big|_1^{\sqrt{2}} = \frac{4\sqrt{2}-5}{3}.$$

(10) 答案: $\frac{3}{2} \ln \frac{\sqrt[3]{4}+1}{2}$

解析: 令 $\sqrt[3]{x}=t$, $x=t^3$, $dx=3t^2dt$, 当 $x=1$ 时 $t=1$, 当 $x=2$ 时 $t=\sqrt[3]{2}$,

$$\text{原式} = \int_1^{\sqrt[3]{2}} \frac{3t^2}{t^3+t} dt = 3 \int_1^{\sqrt[3]{2}} \frac{t}{t^2+1} dt = \frac{3}{2} \int_1^{\sqrt[3]{2}} \frac{1}{t^2+1} (t^2+1)' dt = \frac{3}{2} \ln(t^2+1) \Big|_1^{\sqrt[3]{2}} = \frac{3}{2} \ln \frac{\sqrt[3]{4}+1}{2}.$$

(11) 答案: $\frac{141}{10}$

解析: 令 $\sqrt[3]{x+1}=t$, $x=t^3-1$, $dx=3t^2dt$, 当 $x=0$ 时 $t=1$, 当 $x=7$ 时 $t=2$,

$$\text{原式} = \int_1^2 \frac{t^3-1}{t} \cdot 3t^2 dt = 3 \int_1^2 (t^4-t) dt = 3 \left(\frac{1}{5}t^5 - \frac{1}{2}t^2\right) \Big|_1^2 = \frac{141}{10}.$$

(12) 答案: $1 - \frac{\pi}{4}$

解析: 令 $x = \sin t$, $dx = \cos t dt$, 当 $x = \frac{\sqrt{2}}{2}$ 时 $t = \frac{\pi}{4}$, 当 $x = 1$ 时 $t = \frac{\pi}{2}$,

$$\text{原式} = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos t}{\sin^2 t} \cdot \cos t dt = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^2 t dt = (-\cot t - t) \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} = 1 - \frac{\pi}{4}.$$

(13) 答案: $\sqrt{2} - \frac{2\sqrt{3}}{3}$

解析: 令 $x = \tan t$, $dx = \sec^2 t dt$, $x = 1$ 时 $t = \frac{\pi}{4}$, $x = \sqrt{3}$ 时 $t = \frac{\pi}{3}$, 原式 $= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2 t}{\tan^2 t \cdot \sec t} dt$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos t}{\sin^2 t} dt = -\frac{1}{\sin t} \Big|_{\frac{\pi}{4}}^{\frac{\pi}{3}} = \sqrt{2} - \frac{2\sqrt{3}}{3}.$$

(14) 答案: $\frac{2\ln 2}{3} - \frac{5}{18}$

解析: 原式 $= \int_0^1 \ln(1+x) d\left(\frac{1}{3}x^3\right) = \frac{1}{3}x^3 \ln(1+x) \Big|_0^1 - \frac{1}{3} \int_0^1 \frac{x^3}{1+x} dx$

$$= \frac{1}{3} \ln 2 - \frac{1}{3} \int_0^1 \frac{(x+1)x^2 - (x+1)x + (x+1) - 1}{x+1} dx = \frac{1}{3} \ln 2 - \frac{1}{3} \int_0^1 \left(x^2 - x + 1 - \frac{1}{1+x}\right) dx$$

$$= \frac{1}{3} \ln 2 - \frac{1}{3} \left(\frac{1}{3}x^3 - \frac{1}{2}x^2 + x - \ln|1+x| \right) \Big|_0^1 = \frac{2\ln 2}{3} - \frac{5}{18}.$$

(15) 答案: $\frac{\pi}{4} - \frac{1}{2}$

解析: 原式 $= \frac{1}{2} \int_0^1 \arctan x dx(x^2) = \frac{1}{2} x^2 \arctan x \Big|_0^1 - \frac{1}{2} \int_0^1 \frac{x^2}{1+x^2} dx = \frac{\pi}{8} - \frac{1}{2} \int_0^1 \left(1 - \frac{1}{1+x^2}\right) dx$

$$= \frac{\pi}{8} - \frac{1}{2} (x - \arctan x) \Big|_0^1 = \frac{\pi}{4} - \frac{1}{2}.$$

(16) 答案: 1

解析: 原式 = $\int_0^{e-1} \ln(x+1) d(x+1) = (x+1) \ln(x+1) \Big|_0^{e-1} - \int_0^{e-1} (x+1) \cdot \frac{1}{x+1} dx = e - (e-1) = 1$.

(17) 答案: $\frac{\pi}{2} - \ln 2$

解析: 原式 = $2 \int_0^{\frac{\pi}{4}} x d(\tan x) = 2x \tan x \Big|_0^{\frac{\pi}{4}} - 2 \int_0^{\frac{\pi}{4}} \tan x dx = \frac{\pi}{2} + 2 \ln \cos x \Big|_0^{\frac{\pi}{4}} = \frac{\pi}{2} - \ln 2$.

(18) 答案: $2e^2 + 2$

解析: 令 $\sqrt{x-1} = t$, $x = t^2 + 1$, $dx = 2t dt$, 当 $x = 1$ 时 $t = 0$, 当 $x = 5$ 时 $t = 2$,

原式 = $\int_0^2 e^t \cdot 2t dt = 2 \int_0^2 t d(e^t) = 2te^t \Big|_0^2 - 2 \int_0^2 e^t dt = 4e^2 - 2e^t \Big|_0^2 = 2e^2 + 2$.

(19) 答案: $2\sqrt{2} - 2$

解析: 原式 = $\int_0^{\frac{\pi}{4}} (\cos x - \sin x) dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin x - \cos x) dx = (\sin x + \cos x) \Big|_0^{\frac{\pi}{4}} + (-\cos x - \sin x) \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}$

$= 2\sqrt{2} - 2$.

(20) 答案: $\frac{5}{6}$

解析: 原式 = $\int_{-1}^0 x^2 dx + \int_0^1 x dx = \frac{1}{3} x^3 \Big|_{-1}^0 + \frac{1}{2} x^2 \Big|_0^1 = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$.

2. 答案: $\frac{9}{2} \ln 3 - 2 \ln 2 - \frac{5}{4}$

解析: 令 $t = x - 1$, 则 $x = t + 1$, $f(t) = (t+1) \ln(t+1)$, $\int_1^2 f(t) dt = \int_1^2 (t+1) \ln(t+1) d(t+1)$

$\int_1^2 \ln(t+1) d\left[\frac{1}{2}(t+1)^2\right] = \frac{1}{2}(t+1)^2 \ln(t+1) \Big|_1^2 - \int_1^2 \frac{1}{2}(t+1) dt = \frac{9}{2} \ln 3 - 2 \ln 2 - \frac{5}{4}$.

3. 答案: $\frac{49}{3}$

解析: $\int_0^2 f(x)dx = \int_0^1 (x^2 + 1)dx + \int_1^2 4x^3 dx = \left(\frac{1}{3}x^3 + x\right)\bigg|_0^1 + x^4\bigg|_1^2 = \frac{4}{3} + 15 = \frac{49}{3}.$

4. 答案: 4

解析: 等式两边同时求导得 $f(x) = 2x - \sin 2x$, 再次求导得 $f'(x) = 2 - 2\cos 2x$, 代入 $x = \frac{\pi}{2}$,

$$f'\left(\frac{\pi}{2}\right) = 2 - 2\cos \pi = 4.$$

5. (1) 答案: $\frac{1}{2e}$

解析: 原式 $= \lim_{x \rightarrow 0} \frac{0 - (-\sin x) \cdot e^{-\cos^2 x}}{2x} = \lim_{x \rightarrow 0} \frac{x \cdot e^{-1}}{2x} = \frac{1}{2e}.$

(2) 答案: -2

解析: 原式 $= \lim_{x \rightarrow 0} \frac{0 - 2(\sqrt{1+8x^3} - 1)}{4x^3} = \lim_{x \rightarrow 0} \frac{-2 \cdot \frac{1}{2} \cdot 8x^3}{4x^3} = -2.$

(3) 答案: 0

解析: 原式 $= \lim_{x \rightarrow 0} \frac{\int_0^x (\tan t - t)dt}{x^3 \cdot 1} = \lim_{x \rightarrow 0} \frac{\tan x - x}{3x^2} = \lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{6x} = \lim_{x \rightarrow 0} \frac{\tan^2 x}{6x} = \lim_{x \rightarrow 0} \frac{x^2}{6x} = 0.$

(4) 答案: $\frac{1}{6}$

解析: 原式 $= \lim_{x \rightarrow 0} \frac{\int_0^{\sin x} (e^t - 1 - t)dt}{x^3} = \lim_{x \rightarrow 0} \frac{(e^{\sin x} - 1 - \sin x) \cos x}{3x^2} = \lim_{x \rightarrow 0} \frac{(e^{\sin x} - 1 - \sin x) \cdot 1}{3x^2}$

$$= \lim_{x \rightarrow 0} \frac{e^{\sin x} \cos x - \cos x}{6x} = \lim_{x \rightarrow 0} \frac{(e^{\sin x} - 1) \cos x}{6x} = \lim_{x \rightarrow 0} \frac{\sin x \cdot 1}{6x} = \frac{1}{6}.$$

(5) 答案: 0

$$\text{解析: 原式} = \lim_{x \rightarrow 0} \frac{1 - \cos x^2}{3x^2} = \lim_{x \rightarrow 0} \frac{\frac{1}{2}x^4}{3x^2} = 0.$$

(6) 答案: $-\frac{1}{\pi}$

$$\text{解析: 原式} = \lim_{x \rightarrow 1} \frac{\sin(\pi x)}{-\pi \sin(\pi x)} = -\frac{1}{\pi}.$$

6. 答案: $f(x) = x - \frac{3}{4}$

解析: 设 $A = \int_0^1 f(t) dt$, 则 $f(x) = x + 3A$, 两边同取 $[0, 1]$ 积分, 有 $\int_0^1 f(x) dx = \int_0^1 (x + 3A) dx$, 即

$$A = \left(\frac{1}{2}x^2 + 3Ax \right) \Big|_0^1 = \frac{1}{2} + 3A, \quad A = \frac{1}{2} + 3A, \quad \text{解得 } A = -\frac{1}{4}, \quad \text{所以 } f(x) = x - \frac{3}{4}.$$

7. 答案: 此题错误, 无意义

解析: 此题错误, 无意义

8. 答案: $6 \ln 3 - 4$

$$\text{解析: 原式} = \int_1^9 \ln x d(\sqrt{x}) = \sqrt{x} \ln x \Big|_1^9 - \int_1^9 \sqrt{x} \cdot \frac{1}{x} dx = 3 \ln 9 - \int_1^9 \frac{1}{\sqrt{x}} dx = 6 \ln 3 - 2\sqrt{x} \Big|_1^9 = 6 \ln 3 - 4.$$

9. 答案: $\frac{3\pi}{8}$

$$\text{解析: (法一) 原式} = 2 \int_0^{\frac{\pi}{2}} \sin^4 x dx = 2 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3\pi}{8}.$$

$$\begin{aligned}
 (\text{法二}) \quad \int_0^\pi \sin^4 x dx &= 2 \int_0^{\frac{\pi}{2}} \sin^4 x dx = 2 \int_0^{\frac{\pi}{2}} \left(\frac{1 - \cos 2x}{2} \right)^2 dx = 2 \int_0^{\frac{\pi}{2}} \frac{1 - 2\cos 2x + \cos^2 2x}{4} dx \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(1 - 2\cos 2x + \frac{1 + \cos 4x}{2} \right) dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(\frac{3}{2} - 2\cos 2x + \frac{1}{2} \cos 4x \right) dx \\
 &= \frac{1}{2} \left[\frac{3}{2}x - \sin 2x + \frac{1}{8} \sin 4x \right]_0^{\frac{\pi}{2}} = \frac{1}{2} \left[\left(\frac{3\pi}{4} - 0 + 0 \right) - (0 - 0 + 0) \right] = \frac{3\pi}{8}
 \end{aligned}$$

10. 答案: $e^2 + e - 2$

$$\text{解析: 原式} = \int_0^2 e^{2-x} dx + \int_2^3 e^{x-2} dx = -e^{2-x} \Big|_0^2 + e^{x-2} \Big|_2^3 = -(1 - e^2) + (e - 1) = e^2 + e - 2.$$

11. 答案: $2 - \frac{\pi}{2}$

$$\begin{aligned}
 \text{解析: 原式} &= \int_{-1}^1 \frac{x^3}{1 + \sqrt{1-x^2}} dx + \int_{-1}^1 \frac{x^2}{1 + \sqrt{1-x^2}} dx = 0 + \int_{-1}^1 \frac{x^2(1 - \sqrt{1-x^2})}{x^2} dx = \int_{-1}^1 (1 - \sqrt{1-x^2}) dx \\
 &= 2 \int_0^1 1 dx + 2 \int_0^1 \sqrt{1-x^2} dx = 2x \Big|_0^1 - 2 \cdot \frac{1}{4} \pi \cdot 1^2 = 2 - \frac{\pi}{2}.
 \end{aligned}$$

12. 答案: $4 \sin 1$

解析: 令 $t = \sqrt{x}$, $x = t^2$, $dx = 2t dt$, 当 $x = 0$ 时, $t = 0$; 当 $x = 4$ 时, $t = 2$,

$$\text{原式} = \int_{-1}^1 \cos t \cdot 2(t+1) dt = 2 \int_{-1}^1 t \cos t dt + 2 \int_{-1}^1 \cos t dt = 0 + 2 \sin t \Big|_{-1}^1 = 2(\sin 1 + \sin 1) = 4 \sin 1$$

13. 答案: $\frac{e(\sin 1 - \cos 1) + 1}{2}$

$$\begin{aligned}
 \text{解析: 原式} &= \int_0^1 \sin x d(e^x) = e^x \sin x \Big|_0^1 - \int_0^1 e^x \cos x dx = e \sin 1 - e^x \cos x \Big|_0^1 - \int_0^1 e^x \sin x dx \\
 &= e(\sin 1 - \cos 1) + 1 - \int_0^1 e^x \sin x dx, \text{ 移项得 } 2 \int_0^1 e^x \sin x dx = e \sin 1 - e \cos 1 + 1, \text{ 即原式} \\
 &= \frac{e(\sin 1 - \cos 1) + 1}{2}.
 \end{aligned}$$

14. 答案: 拐点为 $(1, 1 - \frac{2}{e})$

解析: $y' = xe^{-x}$, $y'' = e^{-x}(1-x)$, 令 $y'' = 0$ 得 $x = 1$, 当 $x < 1$ 时 $y'' > 0$ 为凹, $x > 1$ 时 $y'' < 0$ 为

凸, 故 $x = 1$ 处取得拐点, 此时 $y = \int_0^1 te^{-t} dt = \int_0^1 t d(-e^{-t}) = -te^{-t} \Big|_0^1 - \int_0^1 1 \cdot (-e^{-t}) dt = -e^{-1} - e^{-t} \Big|_0^1$
 $= 1 - \frac{2}{e}$, 故拐点为 $(1, 1 - \frac{2}{e})$.

15. 答案: $e + \frac{1}{3}$

解析: 令 $x - 2 = t$, 当 $x = 1$ 时, $t = -1$; 当 $x = 3$ 时, $t = 1$, 原式

$$= \int_{-1}^1 f(t) dt = \left(t + \frac{1}{3} t^3 \right) \Big|_{-1}^0 + e^t \Big|_0^1 = \frac{4}{3} + e - 1 = e + \frac{1}{3}.$$

16. 答案: $-\frac{1}{2}$

解析: 原式 $= \int_1^2 f'(\ln x) d(\ln x) = f(\ln x) \Big|_1^2 = e^{-\ln x} \Big|_1^2 = \frac{1}{x} \Big|_1^2 = -\frac{1}{2}$.

17. 答案: $\frac{1}{(1 + \sin^2 x) \int_0^x \frac{dt}{1 + \sin^2 t}}$

解析: $\frac{dy}{dx} = \left[\ln \left(\int_0^x \frac{1}{1 + \sin^2 t} dt \right) \right]' = \frac{1}{\int_0^x \frac{dt}{1 + \sin^2 t}} \cdot \left(1 \cdot \frac{1}{1 + \sin^2 x} - 0 \right) = \frac{1}{(1 + \sin^2 x) \int_0^x \frac{dt}{1 + \sin^2 t}}.$

四、证明题

1. 答案: 见解析

证明: 令 $F(x) = 3x - 1 - \int_0^x \frac{1}{1+t^4} dt$

$F(x)$ 在 $[0, 1]$ 上连续

$$F(0) = 3 \times 0 - 1 - \int_0^0 \frac{1}{1+t^4} dt = -1 < 0$$

$$F(1) = 2 - \int_0^1 \frac{1}{1+t^4} dt > 0$$

由零点定理可得, $F(x)$ 在 $(0,1)$ 内至少存在一个零点

$$F'(x) = 3 - \frac{1}{1+x^4} > 0, \quad F(x) \text{ 在 } [0,1] \text{ 上单调递增}$$

故 $F(x)$ 在 $(0,1)$ 内有且仅有一个零点, 即 $3x-1-\int_0^x \frac{1}{1+t^4} dt = 0$ 在 $(0,1)$ 内有且仅有一个实根.

2. 答案: 见解析

$$\text{证明: 令 } t = \frac{1}{u}, \text{ 则 } dt = -\frac{1}{u^2} du$$

$$\text{当 } t = x \text{ 时 } u = \frac{1}{x}, \quad t = 1 \text{ 时 } u = 1$$

$$\int_x^1 \frac{dt}{1+t^2} = \int_{\frac{1}{x}}^1 \frac{-\frac{1}{u^2}}{1+\frac{1}{u^2}} du = \int_1^{\frac{1}{x}} \frac{1}{1+u^2} du = \int_1^{\frac{1}{x}} \frac{dt}{1+t^2}, \text{ 得证.}$$

3. 答案: 见解析

$$\text{证明: 令 } t = a+b-x, \text{ 则 } dt = -dx$$

$$\text{当 } x = a \text{ 时 } t = b, \quad x = b \text{ 时 } t = a$$

$$\int_a^b f(a+b-x) dx = \int_b^a f(t)(-dt) = \int_a^b f(t) dt = \int_a^b f(x) dx, \text{ 得证.}$$

4. 答案: 见解析

$$\text{证明: 因为 } f(x) \text{ 是奇函数, 所以 } f(-x) = -f(x)$$

$$\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$$

$$\text{令 } x = -t, \text{ 则 } dx = -dt, \text{ 当 } x = -a \text{ 时 } t = a, \quad x = 0 \text{ 时 } t = 0$$

$$\int_{-a}^0 f(x) dx = \int_a^0 f(-t) \cdot (-1) dt = \int_0^a [-f(t)] dt = -\int_0^a f(x) dx$$

$$\text{代入原式: } \int_{-a}^a f(x) dx = -\int_0^a f(x) dx + \int_0^a f(x) dx = 0 \text{ 得证.}$$